Republic of the Philippines
PARTIDO STATE UNIVERSITY
Camarines Sur

18 October 2021

Dr. Jaime C. Montoya
Executive Director
Philippine Council for Health Research and Development
Department of Science and Technology

Dear **Dr. Montoya**:

The Regional Research Fund (RRF) funded Project entitled "In vitro and In Vivo Anti-Hypercholesterolemic Activity of *Alocasia macrorrhiza* (Araceae) and its Foods Products" has been making great progress since its implementation. However, due to pandemic restrictions, the delivery of chemical reagents and supplies that are very essential for the experimental procedure was delayed. In addition, Food and Drug Administration in Muntinlupa City was struck by fire last August 24, 2021 causing severe damages on their laboratory mice supply which resulted to a delay in the delivery of the procured laboratory animals and making it impossible for the research team to start with the experimental procedures on time.

Moreover, this is to officially request for a three (3) months extension of the duration of the aforementioned research project from its original schedule January 1, 2021- December 31, 2021, to January 1, 2021- March 31, 2022.

Thank you very much.

Sincerely yours,

DARWIN A. GARBELES
Project Leader

Certified availability of Funds:

AMALIA O. OCAMPO
Budget Officer

Recommending Approval:

PATRICIA M. CANDELARIA
VP for Research, Extension, & External Affairs

RAUL G. BRADECINA, Ph.D
SUC President III

Approved:

JAIME C. MONTOYA, Ph.D.
Executive Director
DOST-PCHRD

DOST Form 6

EXECUTIVE SUMMARY FOR THE SEMI-ANNUAL PROGRESS REPORT

Attach DOST Forms 8, 11 and 12)

(1) Program Title: Community Empowerment thru Science and Technology (CEST) Program
Project Title: *In vitro* and *In Vivo* Anti-hypercholesterolemic activity of *Alocasia macrorrhiza* (Araceae) and its Food Products
Project Leader/Gender: Darwin A. Garbeles/ Male
Agency: Partido State University
Address/Telephone/Fax/Email: San Juan Bautista, Goa, Camarines Sur, Region V (Bicol), Philippines

(2) Cooperating Agencies
BCHRD

(3) Site/s of Implementation (Barangay / Municipality / District / Province / Region / Country)
Base Station: San Juan Bautista, Goa, Camarines Sur, Region V (Bicol), Philippines
Site/s of Implementation: San Juan Bautista, Goa, Camarines Sur, Region V (Bicol), Philippines

(4) Project Duration (number of months)	(5) Project Start Date	(6) Project End Date
12 months	01 January 2021	31 December 2021

(7) Major Accomplishments

- Collected *Alocasia macrorrhiza* corms from Brgy. Kinalansan, San Jose, Camarines Sur. The collected plant samples were validated and authenticated by the Department of Biology of the University of the Philippines Baguio.
- Successfully prepared Flour and Starch from *Alocasia macrorrhiza* corms. The prepared flour and starch were processed into gallan flour cookies (GFC) and gallan starch cookies (GSC) as product of this study.
- The produced starch and flour and its processed food products were subjected to nutritional and proximate food analysis.
- Crude extracts from *Alocasia macrorrhiza* corms were collected and purified.
- ELISA microplate reader was procured and tested as a counterpart of the Partido State University.

A. Actual accomplishment of the project (via-a-vis the objectives)

OBJECTIVES	ACCOMPLISHMENTS			
	TARGET	ACTUAL	PERCENTAGE (%)	
			FOR THE PERIOD	CUMULATIVE (FROM START)
The study generally aims to evaluate the anti-hypercholesterolemic activity of the crude extracts from the corms of Alocasia macrorrhiza. Specifically, it aims to: • Determine the phytochemical constituents of the *A. macrorrhiza* corm extracts;	1	0	10%	60%
• Investigate the HMG-CoA reductase percentage of the plant extract using optimized enzymatic kits *In vitro*;	1	0	10%	50%

• Determine the nutritional components of Gallan starch and flour and its food products.	1	1	50%	100%
• Assess the effects of A. macrorrhiza extracts and its food products on blood glucose, total cholesterol, triglycerides, and high-density lipoprotein (HDL), low density lipoprotein (LDL) and very low-density lipoprotein (VLDL)	1	0	10%	50%
• Investigate the histopathological characteristics of the liver treated with different concentrations of A. macrorrhiza extract and its food products.	1	0	10%	50%

B. Catch-up Plan

C. Expected Outputs / 6Ps (Expected Outputs should be measurable.)

	EXPECTED OUTPUTS	ACTUAL OUTPUTS
Publications	1	0
Patents/IP	0	0
Products	1	1
People Services	0	0
Places and Partnerships	0	0
Policy	0	0

(8) Problems/Concerns

- COVID 19 pandemic restrictions
- Laboratory mice facility of the Food and Drugs Administration (FDA) in Alabang was struck by fire

(9) Suggested solutions to the above concerns

- The problems mentioned above mainly contributed to the delay in the conduct of experimental procedures. Thus, extending the duration of the project is suggested.

Prepared by:

DARWIN A. GARBELES
Project Leader

DOST Form A

DEPARTMENT OF SCIENCE AND TECHNOLOGY
Project Line-Item Budget
CY 2021

Project Title: In vitro and In Vivo Anti-hypercholesterolemic activity of *Alocasia macrorrhiza* (Araceae) and its Food Products
Implementing Agency: Partido State University
Project Duration: January to December 2021

ITEM	As Approved **AMOUNT (PhP)** **PCHRD**	1st Re-Alignment **AMOUNT (PhP)** **PCHRD**
I. PERSONNEL SERVICES		
Salary		
1 Project Assistant I @20,109.60 x 13 months	241,315.20	263,216.40
Honoraria		
1 Project leader @8,800/month x 12 months	105,600.00	105,600.00
3 Project Staff Level 1 @4,800/month x 3 months	43,200.00	43,200.00
Subtotal for PS	**390,115.20**	**412,016.40**
II. MAINTENANCE & OTHER OPERATING EXPENSES		
Travel Expenses	30,000.00	20,000.00
Representation Expenses		
Supplies and Materials	150,000.00	154,072.44
Communications	5,000.00	4,860.00
Other Prof. Expenses		
Analytical Services	30,000.00	33,000.00
1 Statistician @3,000/consultation x 1 consultation	6,000.00	3,000.00
Awards/Rewards Expense		
Other MOOE		
Subtotal for MOOE	**221,000.00**	**214,932.44**
III. INDIRECT COST		
Honorarium of Admin Support Staff	30,000.00	30,000.00
Utilities	15,833.64	
Subtotal for IC	**45,833.64**	**30,000.00**
GRAND TOTAL	**656,948.84**	**656,948.84**

Prepared by:

DARWIN A. GARBELES
Project Leader

WORK PLAN

ACTIVITIES	QUARTER	
	4th Q (2021)	1st Q (2022)
1. Delivery of remaining supplies		
2. In Vitro HMG-CoA Inhibitory Assay		
3. In Vivo Assay		
a. Induction of Hypercholesterolemia		
b. Application of Treatments		
c. Termination		
4. Data Analysis and Interpretation		
5. Meetings and Data Presentation		
6. Report Writing		

Prepared by:

DARWIN A.GARBELES
Project Leader